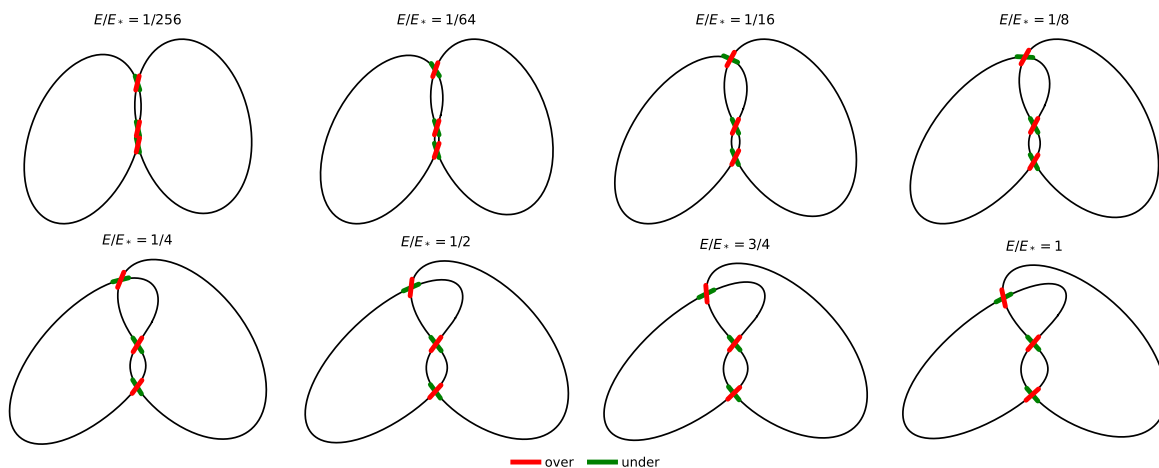




Computed family. Complete stereographic images of K_E , shape-normalized by $R = \sqrt{E}$, with one fixed screen view. Short green/under and red/over segments are centered at the computed crossings and red is always drawn last. The range ends at the first positive ODE singularity $E_* = 4/27$, not at a knot transition.

Model	$C = \{z^2 = w^3\} \subset (\mathbb{C}^2, \omega_0), \quad \omega_0 = \frac{i}{2}(dz \wedge d\bar{z} + dw \wedge d\bar{w}), \quad H = z ^2 + w ^2, \quad K_E = C \cap H^{-1}(E).$
Parametrization	$z = s^3 = u^{3/2}e^{3i\theta}, \quad w = s^2 = ue^{2i\theta}, \quad E = h(u) = u^3 + u^2 = u^2(u+1), \quad u > 0.$
Restricted forms	$\lambda_0 = \frac{1}{2} \operatorname{Im}(\bar{z} dz + \bar{w} dw), \quad \lambda_C = \frac{1}{2}(3u^3 + 2u^2)d\theta, \quad \omega_C = d\lambda_C = \frac{1}{2}(9u^2 + 4u)du \wedge d\theta.$
Time form	$dh = (3u^2 + 2u)du, \quad \omega_C = dh \wedge \eta, \quad \eta = \frac{9u+4}{2(3u+2)}d\theta, \quad \iota_{X_h}\omega_C = -dh, \quad \eta(X_h) = 1.$
Period / action	$T(E) = \oint_{K_E} \eta = \pi \frac{9u+4}{3u+2}, \quad A(E) = \oint_{K_E} \lambda_C = \pi(3u^3 + 2u^2), \quad \frac{dA}{dE} = T(E).$
Energy-native function	$\Phi(E) = 3 - \frac{T(E)}{\pi} = \frac{2}{3u+2}, \quad u = \frac{2(1-\Phi)}{3\Phi}.$
Algebraic elimination	Substituting $E = u^2(u+1)$ gives $(4 - 27E)\Phi^3 - 12\Phi + 8 = 0.$
Direct derivatives	$\Phi_E = -\frac{6}{u(3u+2)^3}, \quad \Phi_{EE} = \frac{12(6u+1)}{u^3(3u+2)^5}.$
ODE	$E(27E - 4)\Phi'' + 2(27E - 1)\Phi' + 6\Phi = 0.$
Gauss form	With $x = \frac{27E}{4}$ and $\Phi(E) = y(x)$, $x(1-x)y'' + \left(\frac{1}{2} - 2x\right)y' - \frac{2}{9}y = 0.$
Solution	$\Phi(E) = {}_2F_1\left(\frac{1}{3}, \frac{2}{3}; \frac{1}{2}; \frac{27E}{4}\right) - \frac{3}{2}\sqrt{E} {}_2F_1\left(\frac{5}{6}, \frac{7}{6}; \frac{3}{2}; \frac{27E}{4}\right).$
Branch sign	For $E > 0$, $u > 0$ and hence $\Phi(E) = 2/(3u+2) < 1$; therefore $\Phi(E) = 1 - \frac{3}{2}\sqrt{E} + O(E)$ and the minus sign is forced.
Integral uniformizer	Only now set $q = \frac{\sqrt{E}}{4}$ and $G(q) = \Phi(16q^2)$. Then
	$G(q) = 1 - 6q + 48q^2 - 420q^3 + 3840q^4 - 36036q^5 + 344064q^6 - 3325608q^7 + \dots$
Coefficients	$G(q) = \sum_{n \geq 0} (-1)^n a_n q^n, \quad a_n = 4^n \binom{3n/2}{n} \in \mathbb{Z}, \quad a_n = [x^n]((1-x)(1-2x))^{-n-1}.$
Recurrence / OEIS	$(n+1)(n+2)a_{n+2} = 12(3n+4)(3n+2)a_n, \quad a_0 = 1, a_1 = 6. \quad \text{Unsigned coefficients: OEIS A244038}. \quad$
First terms	1, 6, 48, 420, 3840, 36036, 344064, 3325608, 32440320, 318704100, ...
Numerical audit	On the original $K_E \subset \mathbb{R}^4$ at eight energies $10^{-4} \leq E \leq 1$ (including $4/27$): constrained-flow quadrature agrees with $T(E)$ to 8.3×10^{-11} relative; an independent polygonal $\oint \lambda_0$ plus centered E -difference gives 1.4×10^{-8} .

Embedded verification payload. Attachments include JSON data, exact SymPy and numerical verifiers, crossing audit, figure code, and this source. They check the forms, cubic/ODE/Gauss pullback, branch sign, integer series/recurrence, numerical integrals, and all eight three-crossing projections. SHA-256: a25dcacbb8100a31...d4ea6b1afa5e23.